

CAMV Sample Midterm 2021

1. Complete the following definition of vectors. Write one word in each of the gaps.

Let A and B are two points. A directed line segment from A to B is a vector. A vector is defined by its direction, and its length.

2. Select the correct answer.

Assume we have the two-dimensional vectors $u = (u_x, u_y)$ and $v = (v_x, v_y)$. In any orthonormal basis, the dot product between u and v can be calculated as

$$A. u \cdot v = \sqrt{(u_x + u_y)^2 + (v_x + v_y)^2}$$

$$B. u \cdot v = \sqrt{(u_x + v_x)^2 + (u_y + v_y)^2}$$

$$C. u \cdot v = u_x u_y + v_x v_y$$

CORRECT ANSWER: D

$$D. u \cdot v = u_x v_x + u_y v_y$$

3. Complete the following statement. Write one word or one number in the gap.

The dot product of two vectors is 0 when the vectors are perpendicular(orthogonal) to each other.

4. Select the correct answer. The cross product of two vectors, $u = (u_x, u_y, u_z)$ and $v = (v_x, v_y, v_z)$ is defined as

- A. a new three-dimensional vector.
- B. a new six-dimensional vector.
- C. a scalar value.
- D. a new two-dimensional vector.

| A. a new three-dimensional vector.

5. Complete the following statement. Write one word in the gap.

| In three dimensions, the length of *cross product* of two vectors, u and v , can be interpreted geometrically as the area of the parallelogram spanned by u and v .

6. Select the correct answer. Which MATLAB function can we use to visualize a vector line as arrow?

- A. plot
- B. quiver
- C. linspace
- D. cross

| B. quiver